An Introduction to Causal Inference

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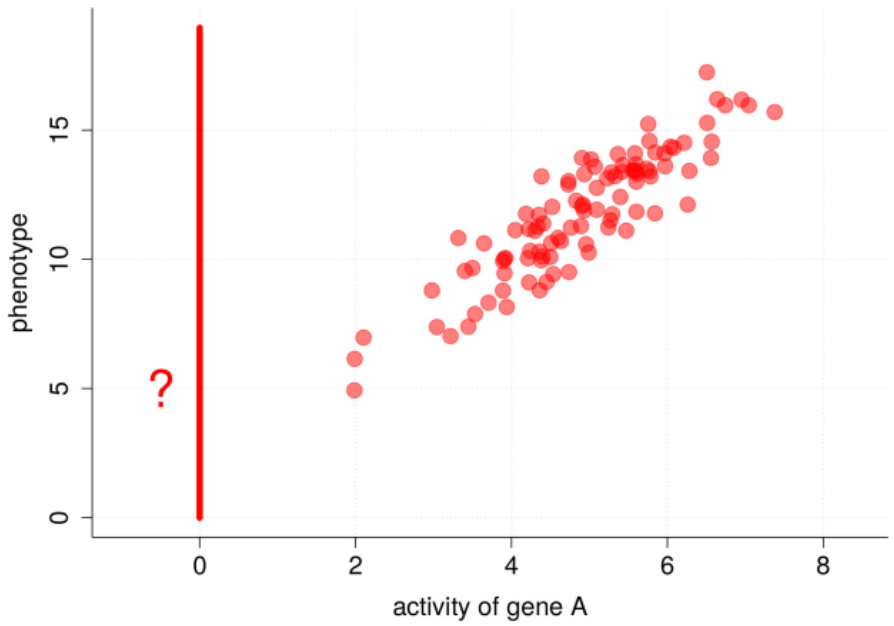
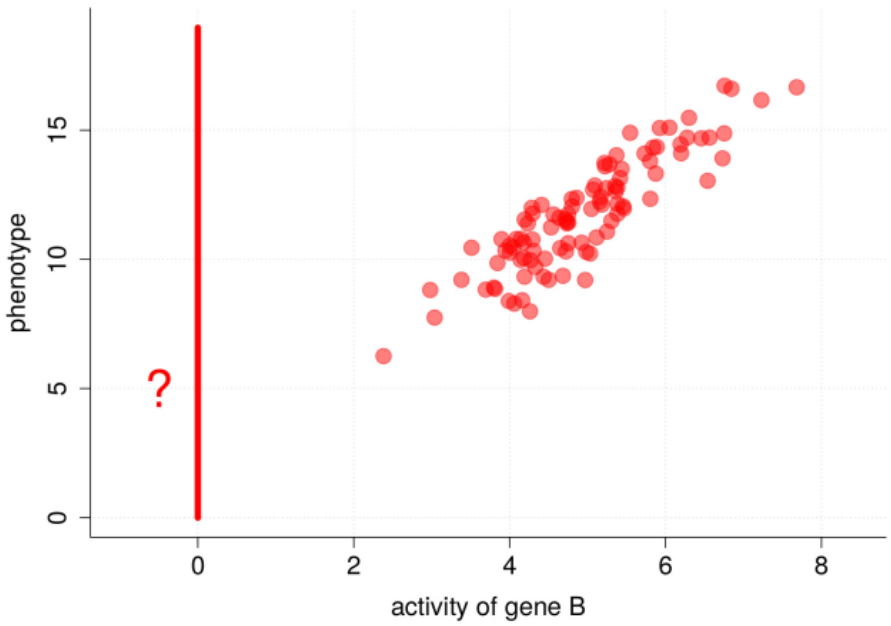
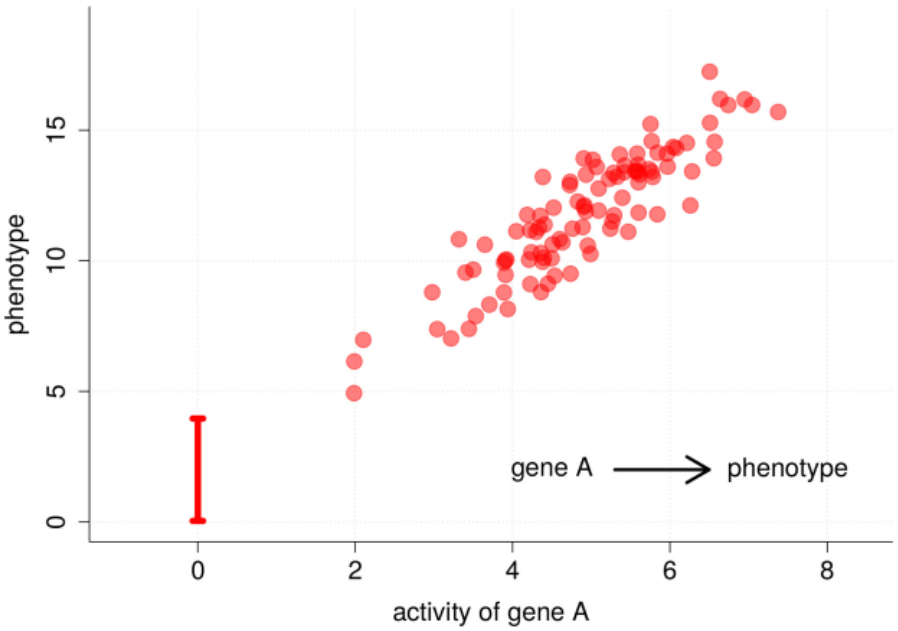
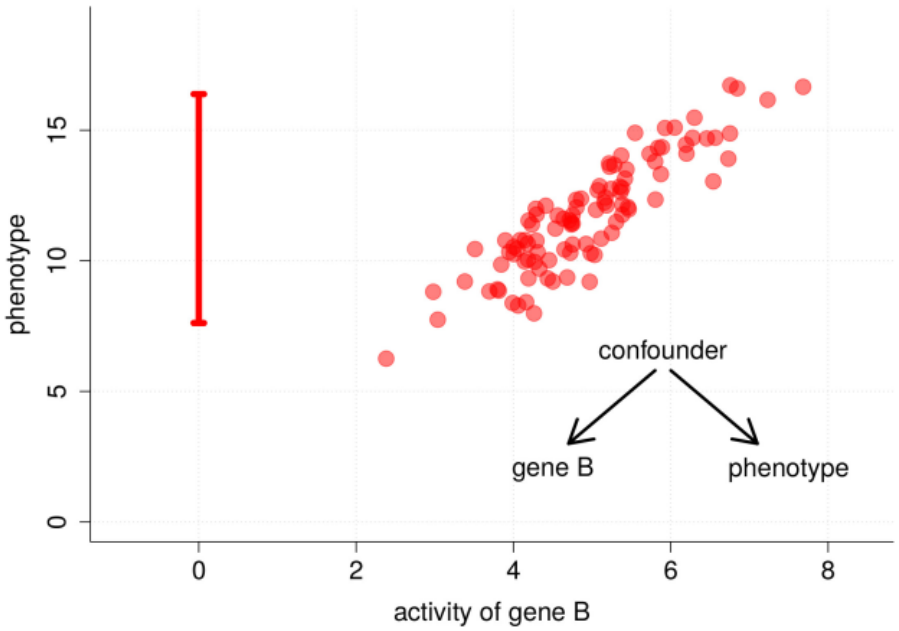
Step 1: Consider the following problem.



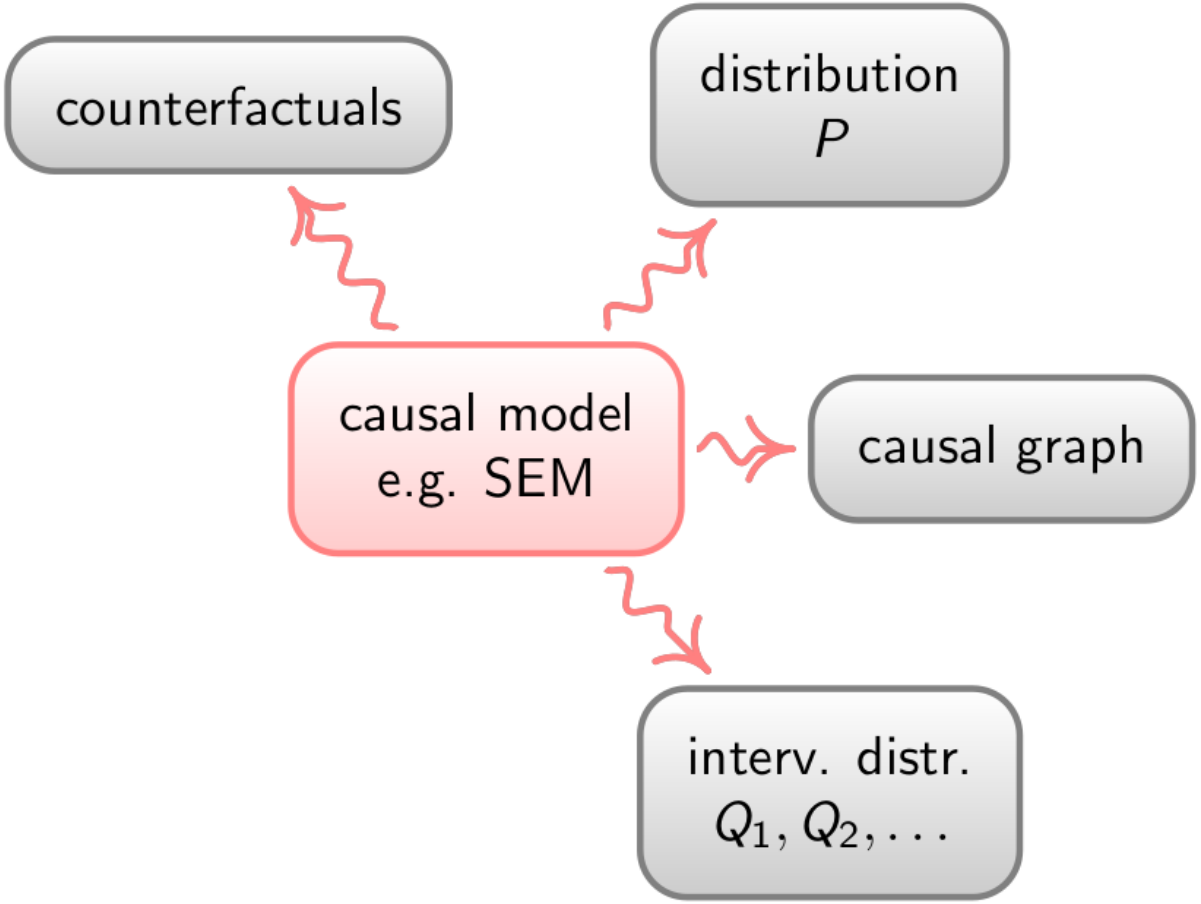


Step 2: Causality Matters.



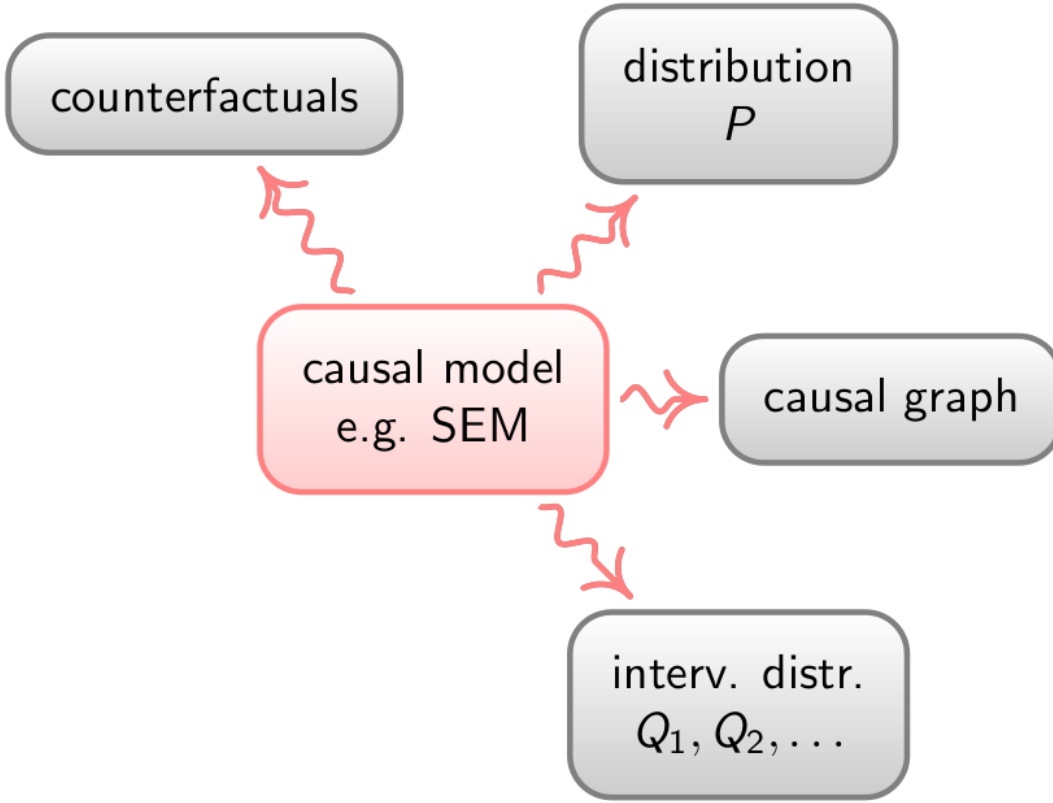


Step 3: Causality Matters.

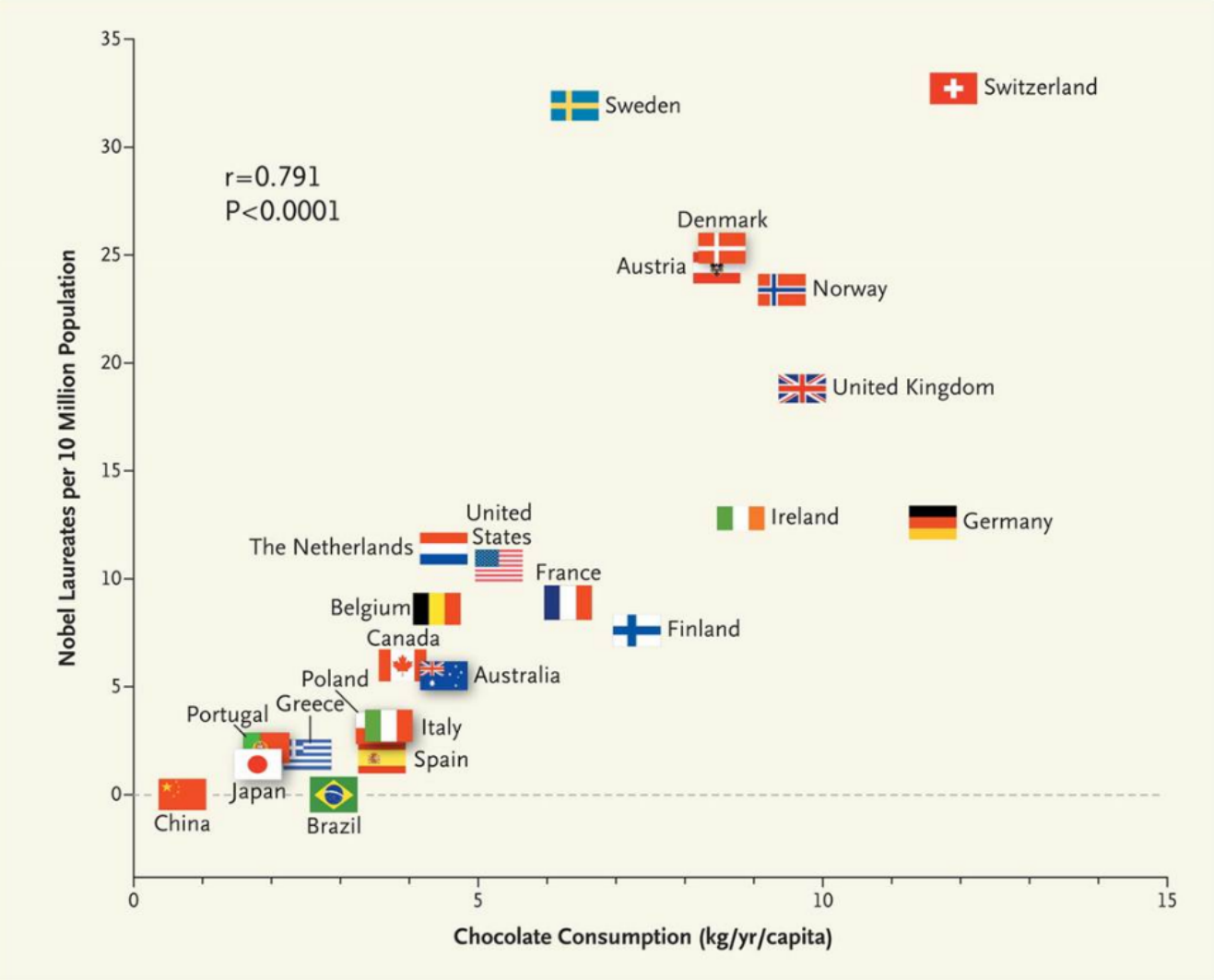


Step 3: Questions

- How to compute interventions without interventions?
- What happens if we have hidden variables?
- What are nice graphical statements?
- Can we infer graph structure from data (or distribution)?



Example: Noble Prize



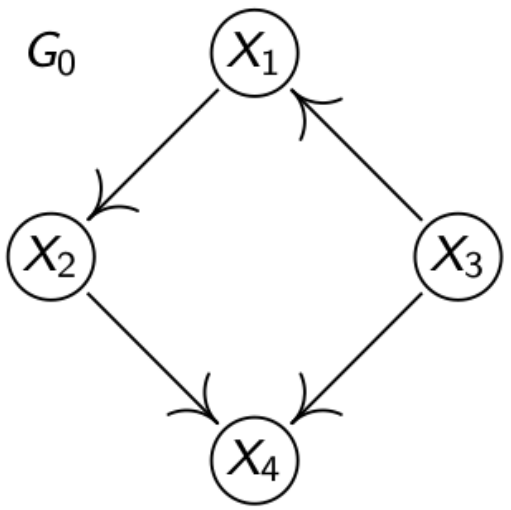
Causal Reasoning

SEM: Structural Equation Model

SEM: Structural Equation Model

$$X_1 := f_1(X_3, N_1)$$
$$X_2 := f_2(X_1, N_2)$$
$$X_3 := f_3(N_3)$$
$$X_4 := f_4(X_2, X_3, N_4)$$

- N_i jointly independent
- G_0 has no cycles



SEM: Structural Equation Model

SEMs can model interventions.

$$X_1 := 0$$
$$X_2 := f_2(X_1, N_2)$$
$$X_3 := f_3(N_3)$$
$$X_4 := f_4(X_2, X_3, N_4)$$

- N_i jointly independent
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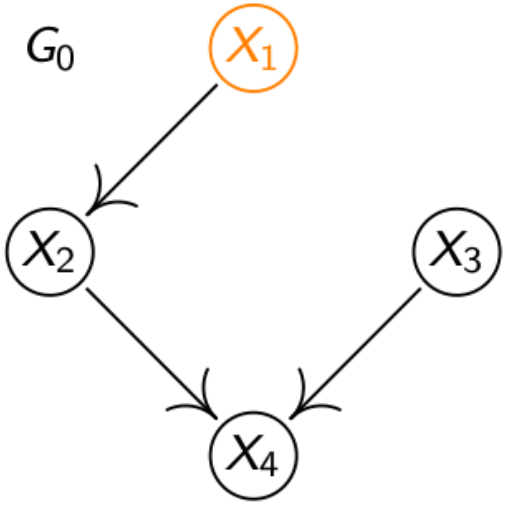


Total Causal Effect

Definition

Given an SEM, there is a total causal effect from X to Y if one of the following equivalent statements is satisfied:

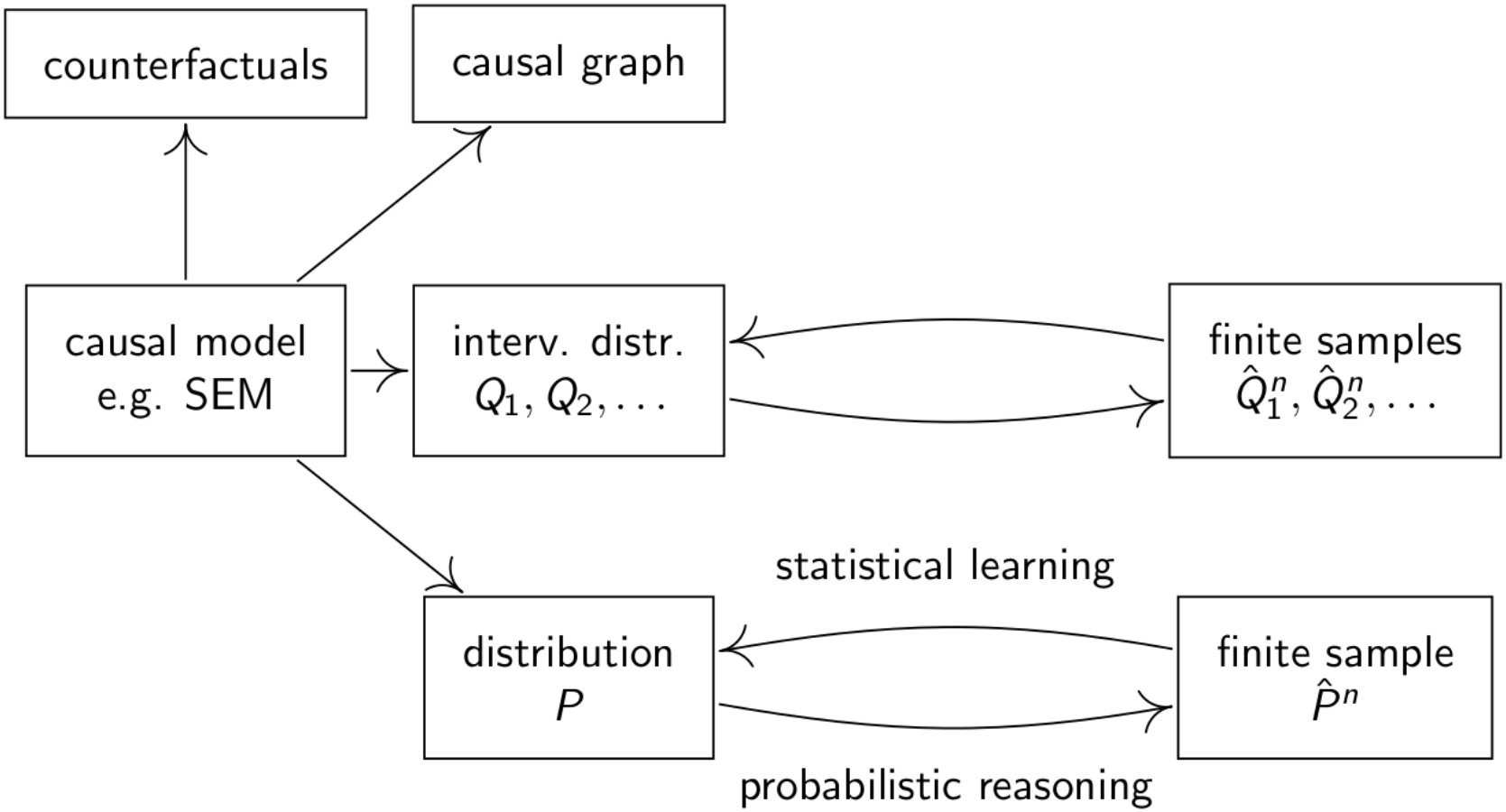
- $X \not\perp\!\!\!\perp Y$ in $\mathrm{P}_{\mathrm{do}(X=\tilde{N}_X)}$ for distribution $\tilde{N}_X$
- There exists $x^{\square}$ such that $\mathrm{P}^{Y}_{\mathrm{do}(X=x^{\square})} \neq \mathrm{P}^{Y}$
- There exists $x^{\square}$ and $x^{\triangle}$ such that $\mathrm{P}^{Y}_{\mathrm{do}(X=x^{\square})} \neq \mathrm{P}^{Y}_{\mathrm{do}(X=x^{\triangle})}$

Quiz

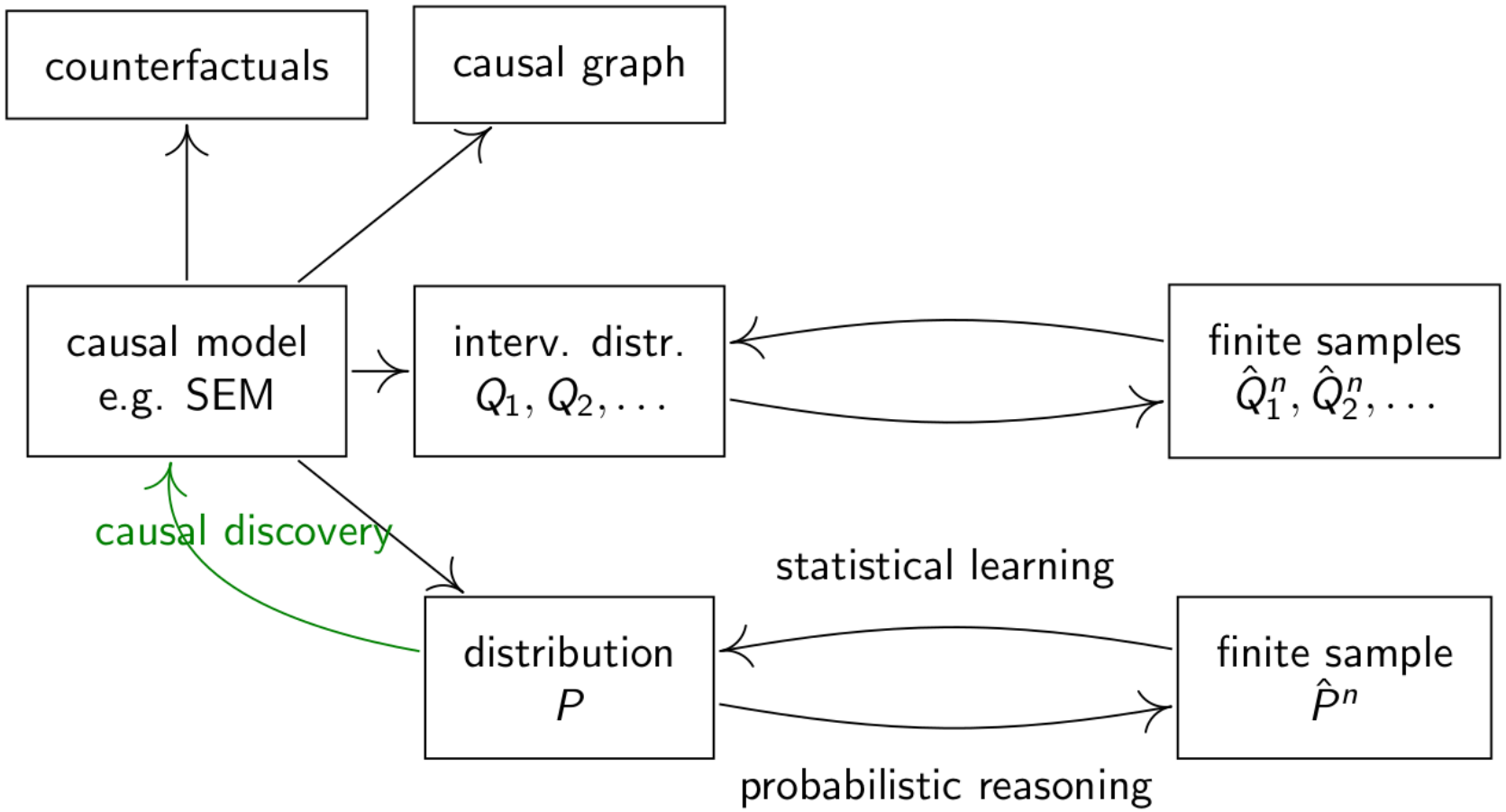
$$P(Y \mid X = x_o) \stackrel{?}{=} P^{Y}_{do(X=x_o)}$$

Causal Discovery

Causal Discovery



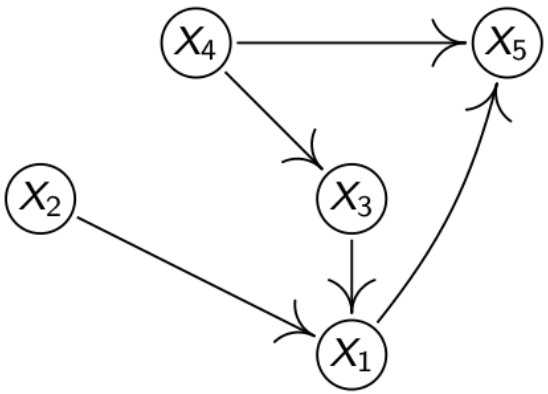
Causal Discovery



What is the relation between P and SEM?

d-separation

X_i and X_j are d-separated by the set S if all paths between X_i and X_j are blocked by S.



$\circ \cdots \rightarrow \circ \rightarrow \cdots \circ$	blocks a path.
$\circ \cdots \leftarrow \circ \rightarrow \cdots \circ$	blocks a path.
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Markov and Faithful

Definition

P is Markov w.r.t. G if
X_i and X_j are d-separated by S in $G \implies X_i \perp\!\!\!\perp X_j | S$

Definition

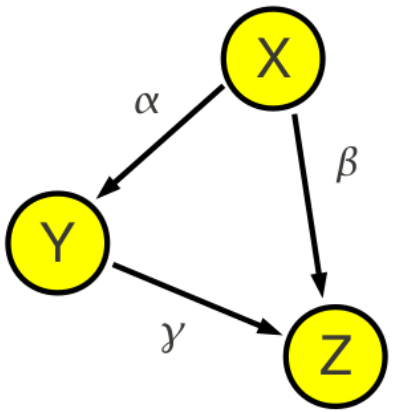
P is faithful w.r.t. G if
$X_i \perp\!\!\!\perp X_j | S \implies X_i$ and X_j are d-separated by S in G

Theorem

Let the distribution P be Markov with respect to a graph. Then the Reichenbach's Common Cause principle is satisfied.

Example of unfaithfulness

$$X = N_x$$
$$Y = \alpha X + N_y$$
$$Z = \beta X + \gamma Y + N_z$$



$\alpha\gamma + \beta = 0$

Unfaithfulness is unlikely because it only occurs for non-generic parameter values.

Idea 1: Conditional Independence-Based Methods

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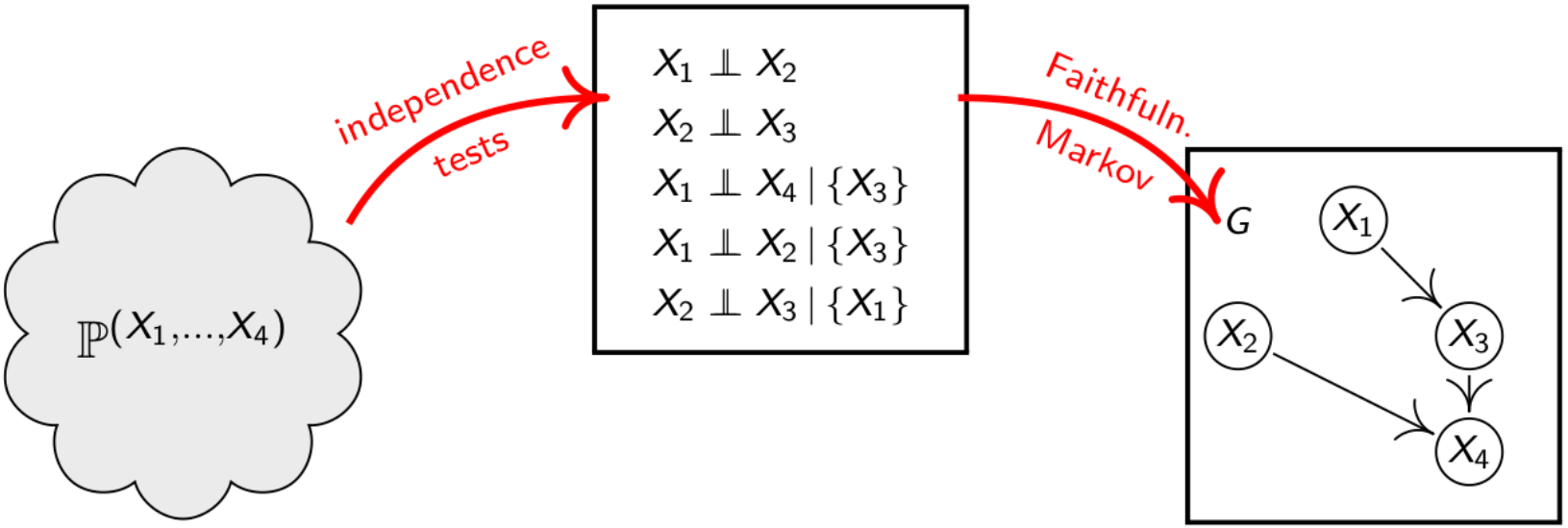

Methods: IC, PC, FCI

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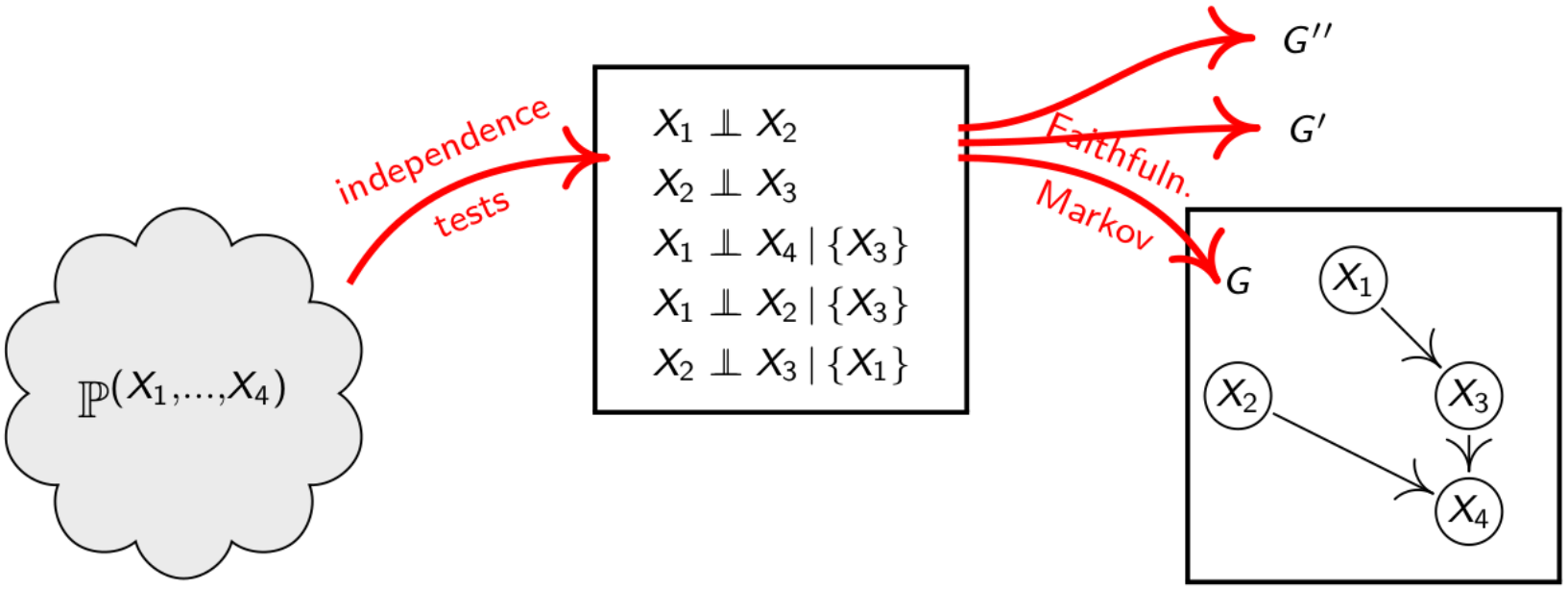
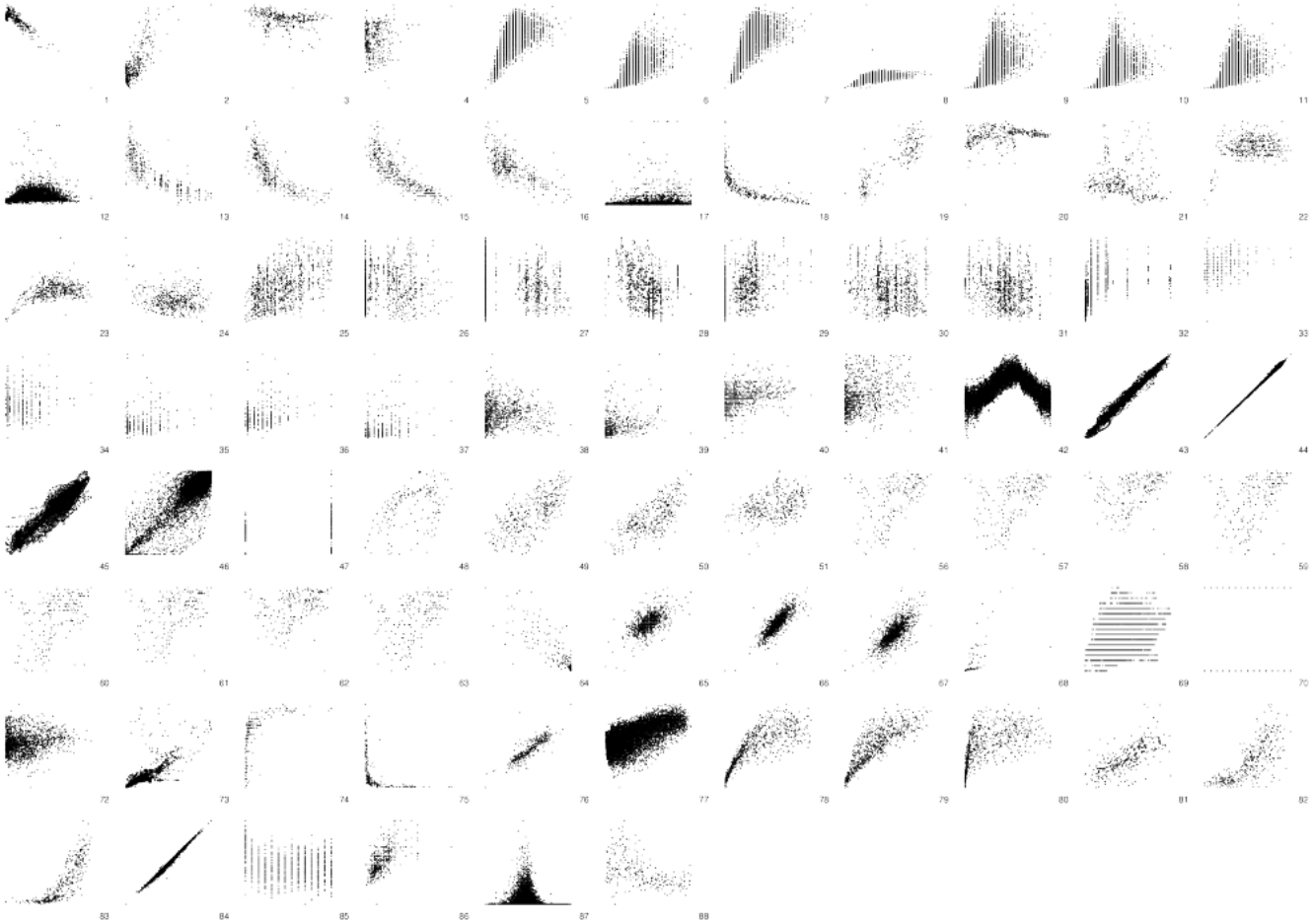


Methods: IC, PC, FCI

Idea 2: Restricted Structural Equation Models

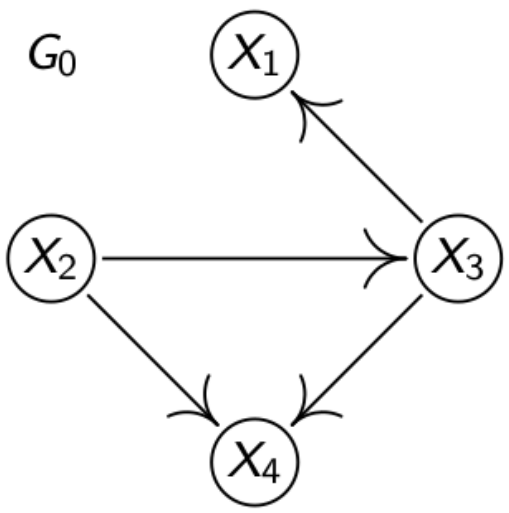
What do we do with two variables? (MPI Cause-Effect Pairs)



Additive Noise Models

$$X_1 = f_1(X_3) + N_1$$
$$X_2 = N_2$$
$$X_3 = f_3(X_2) + N_3$$
$$X_4 = f_4(X_2, X_3) + N_4$$

- $N_i \sim \mathcal{N}(0, \sigma_i^2)$ jointly independent
- G_0 has no cycles



Can the DAG be recovered from $P(X_1, ..., X_4)$? Yes iff f is nonlinear.

Additive Noise Models

Consider a distribution entailed by

$$Y = f(X) + N_Y$$

with $N_Y, X \overset{ind}{\sim} \mathcal{N}$

$$X \longrightarrow Y$$

Then, if f is nonlinear, there is no

$$X = g(Y) + M_X$$

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$$X \longleftarrow Y$$

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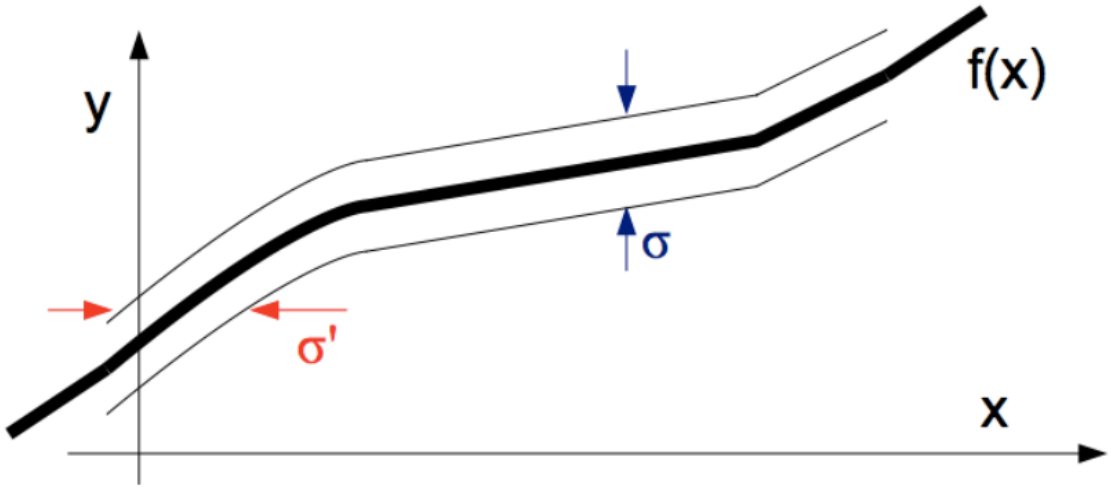
Additive Noise Models

- Assume noise of bounded range
- Additive noise model implies range of Y around f is constant
- For nonlinear f, range of X around backward function non-constant



Information Geometric Causal Inference

if $X \to Y$, the distribution of X and the function f mapping X to Y are independent since they correspond to independent mechanisms of nature.

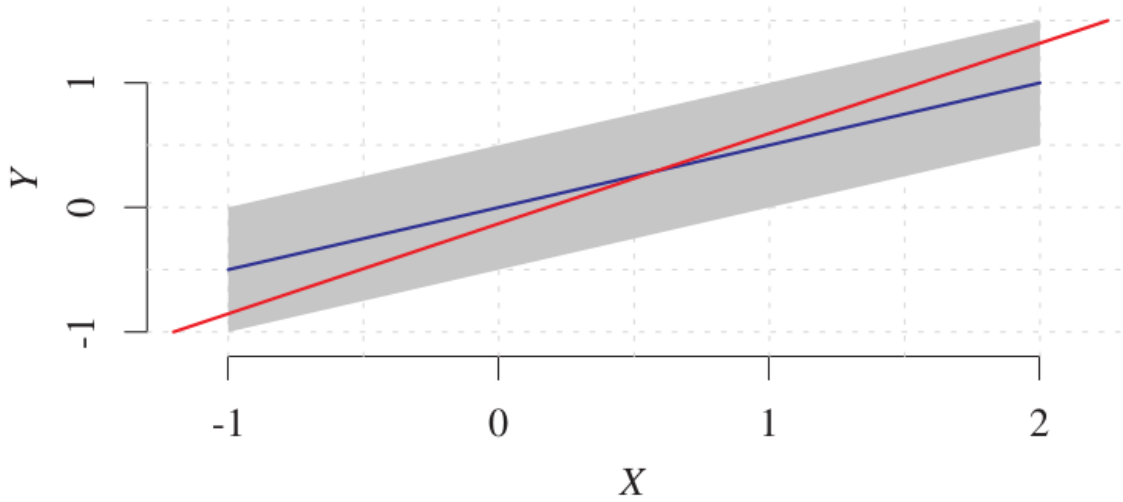
Linear Non-Gaussian Models (LiNGAM)

Theorem

Assume that $P_{X,Y}$ admits the linear model $Y = \alpha X + N_Y, \quad N_y \perp\!\!\!\perp X$, $P_{X,Y}$ admits a backward model iff N_y and X are Gaussian.

Proof.

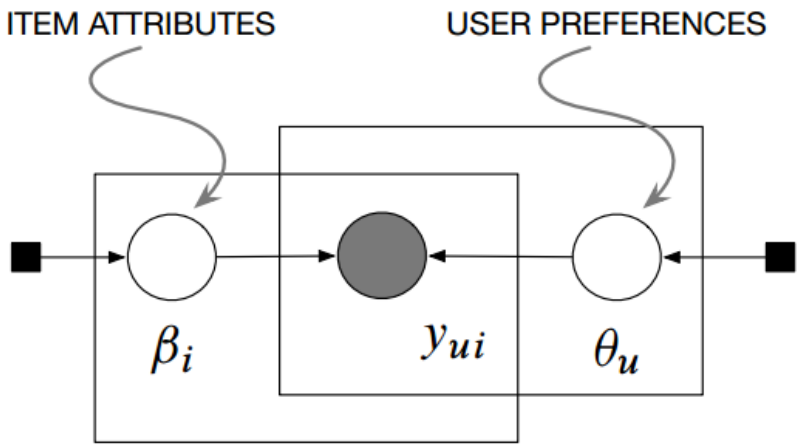
Trivial from Darmois-Skitovic theorem. □



Recommendation Systems

Recommendation Systems

Condition on click data y, estimate $P(\theta, \beta \mid y)$



classical inference of the click model, is incorrect because of the biases induced by the exposure model.